

6.1 Computing the Trade-off Between Decision Loss and Costs of Human Effort

Consider a policy $\tau(\cdot)$ that chooses between full automation (AI), radiologist with access to AI ($E + AI$), or radiologist without access to AI (E), as a function of the AI signal s_i^A . The optimal policy which minimizes the sum of the expected decision-loss (costs of false positives and false negatives) and the monetized time cost of using human radiologists solves:

$$\tau^*(s_i^A) = \arg \min_{\tau \in \{E, E+AI, AI\}} -m \cdot V_{i\tau(s_i^A)} + w \cdot C_{i\tau(s_i^A)}. \quad (8)$$

The first term contains the expected decision-loss from a modality given by $V_{i\tau(s_i^A)} = E \left[V_{ir\tau(s_i^A)} \middle| s_i^A \right]$, which is the expected diagnostic quality averaging over radiologists and cases given the modality, preferences for false positives and false negatives, and the AI signal. Preferences for false positives and false negatives are allowed to vary by pathology but are the same across modalities. We estimate these preferences using data on the binary treatment recommendations of the radiologists in our experiment, given their probability assessments. Details of this preference estimation step are available in appendix C.6. The average cost of false negatives across pathologies is three times the cost of a false positive. Since we do not know the dollar cost of a false negative (m), we will present results for varying values of m .

The second term in the objective function contains $C_{i\tau(s_i^A)} = \left[C_{ir\tau(s_i^A)} \middle| s_i^A \right]$, which is the expected radiologist time for a given modality. If the case is fully automated, this time cost is zero and otherwise those time costs are based on our experimental estimates, which show that radiologists spend more time on cases when presented with AI predictions. For the interpretation of the magnitudes, we translate both radiologists' expected time cost and the expected decision loss into dollars. For the costs of radiologist time, we set $w = \$4$ per minute based on a payment of \$10 per case and an average time per read of 2.5 minutes.

We solve this problem by training a classification tree that assigns the case to a modality conditional on the AI signal $s_{A,i}$. We use 55% of cases as training data for the decision tree, 20% of cases for validation, and 25% in the test set. For this exercise, we focus on two top-level pathologies, cardiomeastinal abnormality, and airspace opacity. Finally, to quantify the drawbacks of biased humans, we will contrast the solution from such a system when the decisions under $E + AI$ are as observed in the data with the system that uses the Bayesian assessments for $E + AI$.

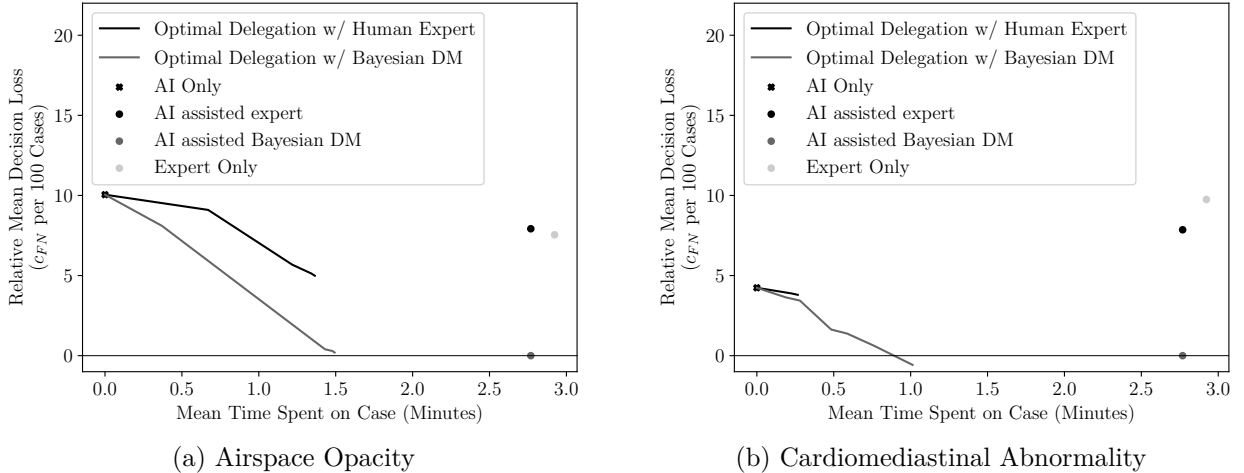
An important restriction in this formulation of the delegation problem is that the signal received by the expert is exogenous and does not respond to the availability of AI. This assumption rules out models of endogenous information acquisition, for example, rational

inattention as in Sims (2003). A potentially interesting aspect is whether a designer can leverage such endogenous responses by designing an information revelation policy that induces effort. We leave such extensions that leverage insights from information design (Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019) to future work, but measure the total amount of time taken by the experts in our experiment with and without AI assistance.

6.2 Results

Figure 8 shows a possibilities frontier for the trade-off between diagnostic quality against decision time for the two top-level pathologies with AI, calculated by varying m . It also shows the corner solutions where all cases are assigned to each of the modalities. The results suggest that there are potential gains from the optimal delegation of cases. An unassisted radiologist takes 2.7 minutes per case, or about \$11, and incurs a relative decision loss of approximately seven for both of the top level pathologies with AI assistance. By moving to a solution on the frontier with a human expert – one whose assessments are as in our experiment – the designer can reduce both decision loss and the time taken per case by delegating many cases to the AI. The frontier with a Bayesian expert, as expected, is much more favorable. Moreover, the frontier for both a Human and a Bayesian expert coincide when the dollar cost of incorrect decisions (m) is small because the vast majority of cases are assigned to the AI.

Figure 8: Loss-Time Frontier

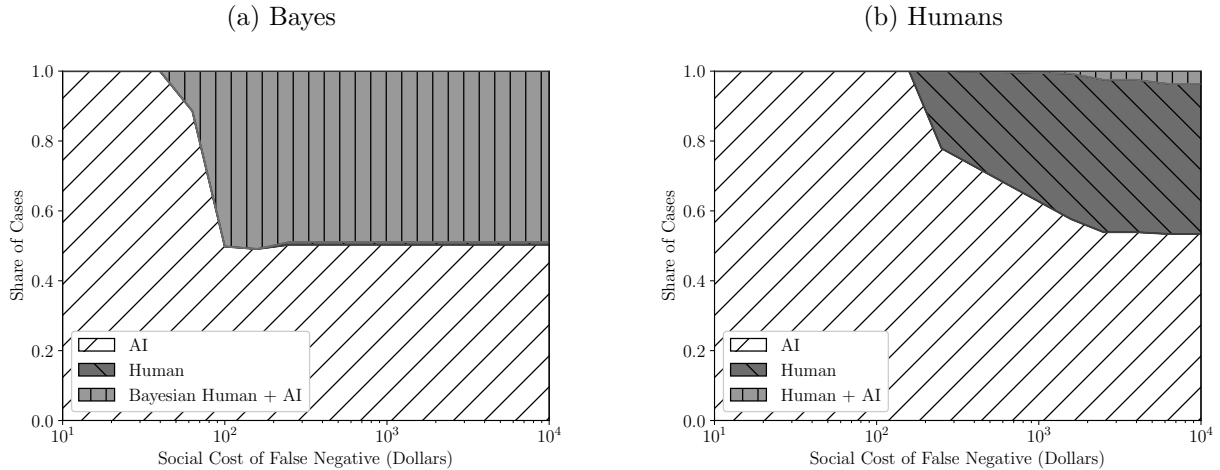


Note: This graph shows how radiologists and the AI perform relative to delegation systems on the frontier of the cost of human time versus decision loss. Panel (a) focuses on airspace opacity. Panel (b) focuses on cardiomedial abnormality.

Next, we investigate the proportion of decisions assigned to the three modalities as we vary

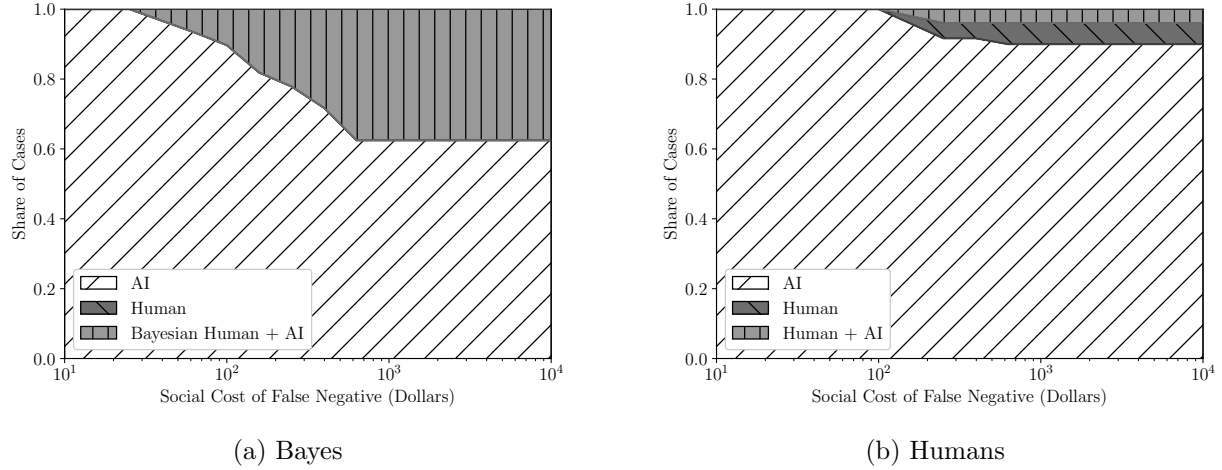
m (figures 9 and 10). Both for the case where radiologists are Bayesian and the observed behavior in our experiment, we find that the AI decides almost all cases if the cost of a false negative – a missed diagnosis – is less than \$100 per case. If radiologists acted as Bayesians with correct beliefs, the share of cases that involve human-AI collaboration rises markedly above a cost of \$100, but even for costs as high as \$10,000, this share does not exceed 50% for airspace opacity and does not exceed 40% for cardiomediastinal abnormality. Moreover, under Bayesian decision-making, the share of cases where only the radiologist decides is negligible for both pathologies and the only reason for using an unassisted radiologist is to save on time costs. When we conduct the same exercise and use the observed behavior of radiologists, we find that humans are involved in approximately 50% of cases for airspace opacity if the cost of a false positive is sufficiently large. For cardiomediastinal abnormality, the share of cases with humans involved only reaches 10% for very large costs of false positives. Moreover, the majority of cases where a radiologist is involved have the radiologist make decisions without AI for both pathologies. A more complete assessment of the optimal combination of human and machine decisions, therefore, confirms the intuition from above that cases are either decided by the radiologist or the AI but not by both of them together.

Figure 9: Airspace Opacity Modality Shares



Note: The graphs show the share of cases decided by each modality (humans, AI, humans+AI) conditional on the cost of a false negative in dollars, denoted m in the text, for airspace opacity. Panel (a) focuses on a Bayesian decision maker. Panel (b) focuses on a human decision-maker with decisions and time-taken as in our experiment.

Figure 10: Cardiomedastinal Abnormality Modality Shares



Note: The graphs show the share of cases decided by each modality (humans, AI, humans+AI) conditional on the cost of a false negative in dollars, denoted m in the text, for cardiomedastinal abnormality. Panel (a) focuses on a Bayesian decision maker. Panel (b) focuses on a human decision-maker with decisions and time-taken as in our experiment.

6.3 Caveats

There are several caveats to the analysis here. The first is that any collaborative system may change radiologists' expectations about the difficulty of cases and adjust strategically to those changing expectations. For example, if radiologists are only handed the most difficult cases, they may exert more effort. We leave such strategic adjustments to a collaborative system for future work. We also have not considered alternative mechanisms to elicit radiologists' signals that may be less prone to errors. For instance, one may want to ask the radiologist to submit their assessment before seeing the AI's assessment and then combine both assessments optimally ex-post. Finally, the solutions presented here treat pathologies as separable. This case is relevant if there is a single pathology of interest.

7 Conclusion

AI is predicted to profoundly reshape the nature of work (see [Felten et al., 2023](#)). Humans are likely to use AI as decision aids for many tasks not only in the long run but also in the medium run for tasks that will ultimately be fully automated. A central question is therefore how humans use AI tools and how tasks should be assigned. To understand the benefits and pitfalls of human-machine collaboration, we conduct an experiment in which radiologists are provided with AI assistance. Besides serving as an iconic example of a